INCH POUND

MIL-T- 38457B (USAF)

23 Apr 90

SUPERSEDING
MIL-T-38457A (USAF)
23 JUNE 1966

MILITARY SPECIFICATION

TRAILER, PALLETIZED CARGO A/M32H-6, 10,000-POUND

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a 10,000-pound capacity, palletized cargo trailer, designated A/M32H-6.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Military

MIL-C-83286	Coating, Urethane, Aliphatic Isocyanate for Aerospace Applications
MIL-P-27443	Pallet, Cargo, Aircraft HCU-6/E
MIL-N-27444	Net, Cargo Tiedown, Pallet HCU-7/E

STANDARDS

Federal

FED-STD-595	Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3920

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Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-281	Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards and Specifications, Order of Preference for the Selection of

OTHER GOVERNMENT DOCUMENTS

Drawings

624002	Trailer, Assembly, Cargo Pallet
	(Copies of drawings are available from WR/ALC/MMDEC, Robins AFB, Ga. 31098-5609)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General requirements. The requirements of DL-624002 and details thereto (see 6.2) shall apply as requirements of this specification.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically-sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Protective treatment. When materials used in the construction of the trailer are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent

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compliance with the performance requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.4.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Recycled materials. All materials, equipment and articles incorporated in the work covered by this specification shall be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.5 Construction. The trailer shall be so constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 Government loaned property. When specified in the contract (see 6.2), the Government will loan the following to the contractor upon his request:

ITEM NUMBER	ITEM DESCRIPTION	QUANTITY
1	Pallet HCU-6/E	1 each
2	Net Tiedown HCU-15/C	1 each
3	Net Tiedown HCU-7/E	2 each

3.6 Performance.

3 6.1 Mobility. The trailer, with rated load in the carrying position, shall be towable at speeds up to 7.5 mph continuously over improved surfaces having abrupt obstructions up to 1 inch in height. The unloaded trailer shall be towable at speeds up to 15 mph continuously over the same surfaces. The trailer with rated load shall withstand sudden stops from a speed of 7.5 mph.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Finishes and protective coatings. Cleaning, painting, and finishing shall be in accordance with MIL-STD-808. The exterior shall be painted with polyurethane paint in accordance with MIL-C-83286, Forest Green, color number 24052 of FED-STD-595.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10 Workmanship. The trailer, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the trailer shall be classified as follows:

a. Preproduction testing (see 4.4).
b. Acceptance tests (see 4.5).

4.3 Test conditions. Servicing or adjustment of any type shall not be permitted during any test specified herein.

4.3.1 Pallet test load. The test load shall consist of a palletized, flexible, evenly distributed gross pallet load of 10,000 pounds with a center of gravity at 48 inches above the pallet. The pallet shall be in accordance with MIL-P-27443. The test load shall be secured to the pallet with a cargo tiedown net in accordance with MIL-N-27444.

4.3.2 Run-in. The nature and extent of the run-in shall be as specified by the manufacturer.

4.4 Preproduction testing (see 6.2).

4.4.1 Test sample. One trailer shall be subjected to the tests specified in 4.4.2.

4.4.2 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of the following:

a. Individual test (see 4.5.1).

4.5.1 Individual tests. Each trailer shall be subjected to an examination of product test as described in 4.6.1.

4.5.2 Rejection and retest. When a vehicle fails to meet the requirement of this specification, vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor and the component that is failed is successfully retested to assure compliance with the specification. Tests not involving the failed component may be continued pending the investigation of a test failure.

4.6 Test methods.

4.6.1 Examination of Product. The trailer shall be thoroughly inspected to determine compliance with the drawings and the requirements of this specification with respect to materials, dimensions, weight, markings, finish, and workmanship. All parts shall function properly and dependably. Observations shall be made to ensure that all accessories such as body, pallet conveyance pallet restraint guides, bumpers, fork lift entries, access walk mountings, running gear, towbar, pintle hook, access walk, reflectors, and tiedown provisions function properly before, during, and after tests.

4.6.2 Overload test. The trailer shall be placed on a smooth, level concrete floor or roadbed. A uniformly distributed load of 25,000 pounds shall be placed on the trailer and held in place until

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inspection of the trailer is accomplished. After the load is removed, any cracks, fractures, or deformation of any part shall be cause for rejection.

4.6.3 Mobility tests

4.6.3.1 Loaded trailer. The trailer, with the test load specified in 4.3.1, shall be towed over improved roads for a distance of 100 miles at an average speed of not less than 6 mph. At the conclusion of each 10 miles, a one hour rest period shall be permitted. During the test, 25 sudden stops, 25 short radius full circle right turns and 25 short full circle left turns shall be accomplished. Except for the specified stops, at no time during the test shall the towing speed be less than 5 mph. During and at the conclusion of the test, the trailer shall be examined and any evidence of permanent deformation, failure, maladjustment or undue wear shall be cause for rejection. Evidence of skewing and overheating during the test shall also be cause for rejection. A temperature condition that will adversely affect normal operation of the trailer shall be considered overheating. The average speed shall be calculated as follows: Distance towed divided by test time (total test time minus one rest period).

4.6.3.2 Empty trailer. The trailer shall be towed at an average speed of not less than 10 mph over improved roads for a distance of 100 miles. At the conclusion of each 10 miles of travel, a 30 minute rest period shall be permitted. During the test, 15 sudden stops, 15 short radius full circle right turns, and 15 short full circle left turns shall be accomplished. Except for the specified stops, at no time during the test shall the towing speed be less than 8 mph by more than 20 percent. At the conclusion of the test, the trailer shall be examined and any evidence of permanent deformation, failure, maladjustment, or undue wear shall be cause for rejection. Evidence of skewing and overheating (see 4.6.3.1) during the test shall be cause for rejection.

4.6.4 Racking tests.

4.6.4.1 Loaded trailer. The trailer, with the test load specified in 4.3.1 in the carrying position, shall be run onto a figure 8 shaped test course. On the long legs of the test course, hardwood obstacles shall be provided consisting of 1- by 3-inch timbers beveled at a 45° angle. The obstacles shall be 36 inches long and shall be placed on each side of the centerline of the course aisle with the longitudinal axis at right angles to the centerline of the course aisle. Distances between the longitudinal axis of the obstacles shall be 60 inches. One set of the obstacles shall be offset by 30 inches from the other set across the centerline of the test course aisle. This layout will allow one wheel at a time to be elevated above grade level. The trailer shall be towed over the obstacle at an average speed of not more than 4 mph for a total of 500 cycles. At the conclusion of each 60 cycles over the obstacles, a 30-minute rest period shall be permitted. At no time during the test shall the towing speed be less than 4 mph. At the conclusion of the test, the trailer shall be examined and any evidence of permanent deformation, failure, maladjustment, overheating (see 4.6.3.1), or undue wear shall be cause for rejection. A cycle of racking shall consist of each wheel of the trailer having passed successively over an obstacle.

4.6.4.2 Empty trailer. The trailer shall be run onto the figure 8 shaped test course described in 4.6.4.1 and shall be subjected to 200 cycles of racking. At the conclusion of the test, the trailer shall be examined for any evidence of permanent deformation, failure, maladjustment, overheating (see 4.6.3.1), or undue wear shall be cause for rejection. The empty trailer shall be towed over the obstacles at an average speed of not less than 10 mph. At no time during the test shall the towing speed be less than 10 mph.

4.6.5 Pallet transfer test. Pallets loaded in accordance with 4.3.1 shall be transferred to and from a conveyorized surface under the following conditions. The horizontal force required to move the pallets shall not exceed 200 pounds. During this transfer, there shall be no contact between the pallets and any part of the trailer, except for the restraint rails and rotating surfaces of the conveyance system. There shall be no permanent deformation or failure.

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a. From the trailer, transfer a pallet along the 88-inch axis to a horizontal conveyorized surface 3 inches below the conveyor system on the trailer.
b. From the horizontal conveyorized surface, transfer a pallet along the 88-inch axis to a trailer conveyance system 3 inches below the conveyorized surface.
c. Steps a. and b. shall be repeated with the transfer along the 108-inch pallet axis.
d. Move a pallet across the trailer so that the pallet strikes the restraints at a speed of 3 mph. This test shall be repeated until the restraints on all four sides of the trailer have been tested with a corner of the pallet striking the center of the stops at a 30° angle.

Steps a through d shall be repeated 50 times.

4.7 Inspection of preparation for delivery. The inspection of the preservation, packaging, packing, and marking shall be in accordance with the requirements of section 5 herein.

5. PACKAGING.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The trailer shall be preserved and packaged in accordance with level A of MIL-STD-281 paragraphs entitled Trailers and trailer dollies.

5.1.2 Level C. The complete trailer, component parts, auxiliary equipment, tools, and technical publications shall be prepared for delivery in accordance with standard commercial practice.

5.2 Packing.

5.2.1 Unless otherwise specified, the trailer shall be shipped as a mobile unit.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The A/M32H-6 trailer is intended for use in air freight terminals and in conjunction with a forklift for aircraft loading and unloading operations. The trailer is equipped with running gear and incorporates a towbar and pintle for movement by a tow vehicle in single or train fashion. The trailer is intended for carrying palletized loads of 10,000 pounds over medium distances on improved road surfaces.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. In the event it becomes necessary to change the drawing (see 3.2) in order to achieve the performance requirements of this specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required perform ance. To the extent approved, such changes will be processed in accordance with the changes clause of this contract.
c. When required, pallets and nets will be loaned by the Government to the contractor (see 3.5. 1).
d. Location and conditions for preproduction testing (see 4.4).

6.3 Subject term (key word) listing.

Conveyance System
Pallet
Towable

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force - 99

Reviewer:
Air Force - 85

Preparing activity:
Air Force - 84

Project No:
3920-F184

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